Python(22AIE205) Programming Lab

Weekly Lab Report-3

LAB 3: Control statements.

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Experiment Record:

Date:	25-07-2026
Experiment Name:	Control statements
Aim:	To practice Control statements using Python IDLE.
Software Required:	Python 3.x and Python IDLE

Questions Given:

1. 1. Write a Python program using a while loop to simulate a bus ticket booking counter with the following requirements:
    - The bus has a total capacity of **40 seats**.
    - The program should keep asking the user how many tickets they want to book, until they choose to exit (by entering 0).
    - For each booking request:

- If the requested number of tickets is **available**, confirm the booking and reduce the available seat count.
- If the requested number **exceeds available seats**, display an appropriate message showing how many seats are actually left.
- If the bus becomes **fully booked**, display a "Bus Full" message and stop accepting further bookings automatically.

- At the end (whether by user exit or bus becoming full), display the **total number of tickets sold** and the **total seats remaining**.

2. Write a program using a while loop that allows a user a maximum of 3 attempts to enter the correct 4-digit ATM PIN. Display "Access Granted" on success. If all 3 attempts fail, display "Account Locked" and terminate.

3. Write a program using a while loop that generates a random number between 1 and 50 and allows the user to keep guessing until they guess correctly. After each guess, display whether the guess was too high or too low. Display the total number of attempts taken once the correct number is guessed.

4: Write a program using a while loop to manage a shop's inventory starting with 50 units in stock. The program should repeatedly accept the number of units sold, update the stock, and display a "Reorder Alert" when stock falls to 10 or below. The loop should terminate when the user enters -1 or stock reaches 0.

Programs, Sample Outputs and Results:

Question 1: Write a Python program using a while loop to simulate a bus ticket booking

Program:

```
File  Edit  Format  Run  Options  Window  Help
total_seats=40
available=total_seats
sold=0
print("The Available Seats in Bus:",available)
while available>0:
    tickets=int(input("Enter Number Of tickets to book:"))
    if tickets==0:
        print("Exiting Booking System")
    elif tickets<=available:
        available-=tickets
        sold+=tickets
        print(f"Your Booking Confirmed by booking-{tickets} and seats left {available}")
    else:
        print(f"Cannot book {tickets} only left seats are {available}")
    if available==0:
        print("Bus is Full No More Booking!")

    print("The number of seats booked:",sold)
    print("The number of tickets available:",available)
```

Input and Output:

```
>>>
======= RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\bus.py =======
The Available Seats in Bus: 40
Enter Number Of tickets to book:12
Your Booking Confirmed by booking-12 and seats left 28
The number of seats booked: 12
The number of tickets available: 28
Enter Number Of tickets to book:
```

Result: The program executed successfully by taking input from user to book bus tickets and displayed number of tickets available after booking.

Question-2: Write a program using a while loop that allows a user a maximum of 3 attempts to enter the correct 4-digit ATM PIN. Display "Access Granted" on success. If all 3 attempts fail, display "Account Locked" and terminate.Program:

Python Code:

```
correct=9366
attempts=0
while attempts<3:
    pin=int(input("Enter You 4 digit AIM pin:"))
    if pin==correct:
        print("Access Granted For Transaction!")
    else:
        attempts += 1
        print("wrong pin,you have only left with chances",3-attempts)
```

Input&Output:

```
======= RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\atm.p
Enter You 4 digit AIM pin:1234
wrong pin,you have only left with chances 2
Enter You 4 digit AIM pin:2347
wrong pin,you have only left with chances 1
Enter You 4 digit AIM pin:8763
wrong pin,you have only left with chances 0
```

Result:The program is executed successfully ,by taking input from user and applied all conditions by given 3 chances for correct input identification

Question-3: Write a program using a while loop that generates a random number between 1 and 50 and allows the user to keep guessing until they guess correctly. After each guess, display whether the guess was too high or too low. Display the total number of attempts taken once the correct number is guessed.

Program:

```
import random
target = random.randint(1, 50)
print(f"(Debug) Target number is: {target}")

guess = 0
attempts = 0
while guess != target:
    guess = int(input("Guess the number (1-50): "))
    attempts += 1
    if guess < target:
        print("Too low!")
    elif guess > target:
        print("Too high!")
    else:
        print("Correct!")
print(f"You guessed it in {attempts} attempts.")
```

Input & Output:

```
>>>
====== RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\game1.py ===
(Debug) Target number is: 25
Guess the number (1-50): 20
Too low!
Guess the number (1-50): 30
Too high!
Guess the number (1-50): 25
Correct!
You guessed it in 3 attempts.
>>>
```

Question-4: **Write a program using a while loop to manage a shop's inventory starting with 50 units in stock. The program should repeatedly accept the number of units sold, update the stock, and display a "Reorder Alert" when stock falls to 10 or below. The loop should terminate when the user enters -1 or stock reaches 0.**

Program:

```
stock = 50

while stock > 0:
    sold = int(input("Enter the number of units sold (-1 to stop): "))

    if sold == -1:
        print("Inventory management stopped.")
        break

    if sold > stock:
        print("Not enough stock available.")
        continue

    stock = stock - sold
    print("Available stock:", stock)

    if stock <= 10 and stock > 0:
        print("Reorder Alert")

if stock == 0:
    print("Stock is empty.")
```

Input&Output:

```
====== RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\stock.py ====
Enter the number of units sold (-1 to stop): 35
Available stock: 15
Enter the number of units sold (-1 to stop):
```

4. Observations

Imported “random” ,and used randint() function to generate a random integer.

And loop exits when condition fails or break.

5. Overall Result

All 4 programs Executed Successfully used while and if elif in necessary conditions.